

AI Based Anti-Proxy Attendance System Using Facial Recognition

Final Year Project Proposal (BSCS)

By

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Abstract

Educational institutions are beginning to embrace smart technologies in recent years to enhance classroom management and enhance the efficiency of the educational process. Manual roll calls and other traditional attendance systems take up valuable lecture time, and can be prone to human error and proxy attendance.

These attendance problems lead to difficulties in keeping accurate students' records and less productivity in class. While digital attendance systems are available, many of them simply make record keeping easier, and do not offer adequate security protocols to stop fake attendance.

The proposed work is an AI-Based Anti-Proxy Smart Attendance System Using Facial Recognition, a modern web-based system to automatically record the attendance of students in classroom using the camera. The system will use face detection, live face verification technology including blinking and head movements to ensure that only physically present students are marked as 'present'.

The proposed architecture will be a modern full-stack system comprising the frontend with React.js, backend with Node.js/Express.js, and AI with Python/OpenCV.

The primary objective of this project is to design a secure, intelligent and user friendly attendance system to overcome the drawbacks of existing system such as inaccurate attendance, saving the time of the teacher in the classroom and preventing proxy attendance.

Introduction

Managing attendance is a crucial aspect of school or college. In schools/colleges/universities, teachers typically have to take attendance by hand, particularly in large classes for a few minutes. This is a traditional way which wastes lecture time and may cause errors in attendance records.

Another big problem is proxy attendance: students sign in for fellow students who are not in attendance. There are attendance systems available, but many don't check to see if the student is in the classroom. The development of Artificial Intelligence, Computer Vision and the current web technologies have made it possible to develop smart attendance systems that can now automatically recognize students through the camera. Facial recognition can be combined with other verification techniques such as live verification, making attendance systems more accurate and secure.

The aim of this project is to create a smart attendance system on Web which automatically registers attendance and will also not allow attendance to be falsified using anti-proxy verification methods.

Problem statement

There are a number of attendance management challenges in educational institutions. Time is being still wasted and management is difficult in many classrooms, where students are still manually marked.

Key issues are:

Manual attendance takes lecture time.

Errors in attendance records that are human related.

Students taking proxy attendance

The current systems which do not have live verification.

Inability to handle attendance information effectively.

In classrooms:

Teachers waste precious teaching time.

Record keeping may be inaccurate

Fake attendance has an impact on transparency and discipline.

Large classroom is hard to control.

Considering the above, there is a demand for a solution to automatically record students' attendance and also to ensure their presence in real time in a secure and intelligent manner.

Prospective Users

The proposed system can be useful for various users such as:

Schools/colleges for school attendance

Universities with a significant number of students

Those teachers who desire automated attendance management

Educational administrators who are keeping attendance records.

Training institutes and academies

The system is planned to be modern, efficient and user friendly to make it easily adoptable by institutions.

Literature review

Today, a lot of attendance systems are based on RFID cards, fingerprint scanners, or manual entries into a digital register. Although these systems are better for storing attendance, they have a number of drawbacks.

In recent years, facial recognition attendance systems have also been created. These systems are based on computer vision techniques for identification of students using camera. But several of the current systems:

Do not use face recognition with the live verification.

Can be fooled using photos or videos

Focus mainly on attendance marking only

Don't have adequate web dashboards and analytics

Works have been done on anti-spoofing and live face verification techniques, but many of the solutions are costly or are hard to use in education institutions.

Lack of systems that combine:

Real-time facial recognition

Anti-proxy verification

Attendance analytics

Modern web-based dashboards

User-friendly interfaces

The objective of this project is to bridge this gap by creating a secure, intelligent and complete attendance management system.

Project scope

The scope of this project is to design and develop a web-based smart attendance system that automatically records student attendance using classroom cameras.

- Use classroom cameras to automatically record student attendance.
- Identify students using facial recognition technology.
- Real time detection and recognition of students
- Replace manual attendance with electronic attendance.
- Use anti-proxy verification based on blink and face movements detection
- Provide secure and accurate attendance marking.
- Create an easy-to-use dashboard to monitor attendance.
- Generate attendance reports and analytics.
- Record students' absences and attendance.
- Support export of attendance to Excel/CSV.
- Efficient Use of Classroom Time, Attendance Management.

- Develop a dependable and smart attendance solution for schools.

Software Components:

Modern full-stack technology will be used to develop the System.

- Frontend
- React.js

React.js will be used to design a responsive and interactive web interface for:

- Login system
- Student registration
- Live attendance dashboard
- Reports and analytics
- Backend
- Node.js and Express.js

The back end will take care of:

- API development
- Attendance processing
- To support data communication and authentication. To facilitate data communication and authentication.
- Database operations
- Database
- MongoDB

MongoDB will store:

- Student records
- Attendance history
- User login information
- Attendance analytics
- Developing AI and Computer Vision Module
- Python + OpenCV

Python and OpenCV will be used for:

- Face detection
- Facial recognition
- Blink detection
- Live face verification
- Anti-proxy mechanisms
- Real-Time Communication
- Socket.io

Socket.io will provide:

- Real-time attendance updates
- Live notifications
- Dynamic dashboard refresh

Project development methodology

The use of Prototype Based Development Approach will enable us to develop the system step by step, continuously improving it.

Development Steps

- Gathering and planning the requirements.
- Crafting system design and web interface. Creating system architecture and web interface.
- Building frontend modules with React – the front-end part of a web app.
- Building back-end APIs with Node.js
- Using Python with OpenCV to implement facial recognition.
- Integrating MongoDB database
- Testing the entire system and enhancing accuracy

This way, problems will be discovered early on and the quality of the system will be ensured.

Tools and Technologies:

The following tools and technologies will be used:

- React.js
- Node.js
- Express.js
- MongoDB
- Python
- OpenCV
- Facial Recognition Libraries
- Socket.io
- NumPy and Pandas
- Machine Learning techniques for face verification

Project milestones and deliverables

The project will be completed within five months.

Gantt Chart :

1	Project Timeline					For scheduling tasks and tracking progress	
2	Tasks	Week 1	Week 2	Week 3	Week 4	Week 5	
3	Proposal Writing						
4	System Design						
5	Frontend Development						
6	Backend Development						
7	AI Implementation						
8	System Integration						
9	Testing						
10	Documentation						
11	Final Presentation						

Deliverables:

At the end of the project, we will provide:

- Trusted, comprehensive web based attendance solution.
- AI-based facial recognition module
- Anti-proxy verification system
- Interactive React dashboard
- Attendance reports and analytics.
- Final project documentation
- The presentation slides and system demonstration.

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